IOT & DIGITAL DEVICE FORENSICS

Weekly task list

A fictional sensor export combines device inventory, telemetry and gateway events. Work only with supplied files; no hardware access, radio capture or device modification is needed.

The investigation question

How do device clock assumptions and telemetry quality affect the gateway chronology?

Your four weekly tasks

Week 1 | Device ecosystem mapping

Inventory devices.csv, telemetry.csv and gateway.csv; compute file hashes and assign evidence IDs. Draw device-to-gateway relationships and list the device clock assumptions and measurement units.

Output: ecosystem_map.pdf, evidence_register.csv and preservation_plan.pdf

Week 2 | Telemetry quality review

Group readings by device and compute count, minimum, maximum and time gaps. Identify duplicate sequence numbers, missing intervals and the deliberately offset sensor clock.

Output: telemetry_summary.csv, quality_flags.csv and reasoning.pdf

Week 3 | Gateway and device correlation

Normalize time using a clearly stated clock-offset assumption; preserve raw device timestamps. Correlate gateway reconnect events with telemetry gaps and flag matches that depend on an assumption.

Output: iot_timeline.csv, clock_assumptions.pdf and correlation_table.csv

Week 4 | IoT case report

Prepare a 4-6 page report with ecosystem diagram, telemetry quality results and correlated gateway chronology. Explain the impact of clock differences, absent firmware history and unavailable physical-device context.

Output: iot_report.pdf and iot_appendix.zip

Read pages 2-5 for the full checklists. Submission rules follow on page 6; requirements and installation are on pages 7-8. Worked methods and three reading downloads are on page 9. The complete roadmap is the final page.

WEEK 1 | TASK INSTRUCTIONS

Device ecosystem mapping

Objective

Create a trustworthy baseline before interpreting any record.

Do this in order

1. Inventory devices.csv, telemetry.csv and gateway.csv; compute file hashes and assign evidence IDs.

2. Draw device-to-gateway relationships and list the device clock assumptions and measurement units.

3. Describe an appropriate preservation plan for a real device, while stating no actual hardware acquisition occurs in this exercise.

4. Create a case note naming devices.csv, telemetry.csv and gateway.csv. For each, record evidence ID, original filename, byte size, SHA-256 and where the PFSA ZIP was downloaded. Keep the supplied manifest with the ZIP.

5. Group readings by device before checking sequence values. A repeated sequence across different devices is not necessarily a duplicate.

6. Recheck one result independently: return to the source, repeat the calculation or lookup, and record whether it agrees. Keep a short correction log instead of silently revising previous findings.

Required hand-in

ecosystem_map.pdf, evidence_register.csv and preservation_plan.pdf

Name the upload PFSA_<student-id>_iot_W1.pdf or .zip. A ZIP may contain the report plus CSV, SQL, images or other permitted supporting work. Do not rename a non-PDF file to .pdf. Keep the one portal upload below 10 MB.

Portal description and acceptance check

Write at least 200 genuine words in the description box: what you examined, how you processed it, exact evidence references, results, uncertainty and what you learned. Before submitting, open the uploaded work locally, confirm all filenames resolve, and compare original-file hashes with your intake record.

WEEK 2 | TASK INSTRUCTIONS

Telemetry quality review

Objective

Turn source records into a documented examination worksheet.

Do this in order

1. Group readings by device and compute count, minimum, maximum and time gaps.

2. Identify duplicate sequence numbers, missing intervals and the deliberately offset sensor clock.

3. Decide whether each anomaly may reflect collection loss, clock drift or a device event; keep observations and hypotheses separate.

4. Keep all original columns from devices.csv, telemetry.csv and gateway.csv. Add new analysis columns rather than replacing source identifiers or timestamps. Preserve the query, formula or filter used.

5. Calculate reading count, min/max and consecutive time gaps separately. Do not insert invented readings into the original evidence to fill gaps.

6. Recheck one result independently: return to the source, repeat the calculation or lookup, and record whether it agrees. Keep a short correction log instead of silently revising previous findings.

Required hand-in

telemetry_summary.csv, quality_flags.csv and reasoning.pdf

Name the upload PFSA_<student-id>_iot_W2.pdf or .zip. A ZIP may contain the report plus CSV, SQL, images or other permitted supporting work. Do not rename a non-PDF file to .pdf. Keep the one portal upload below 10 MB.

Portal description and acceptance check

Write at least 200 genuine words in the description box: what you examined, how you processed it, exact evidence references, results, uncertainty and what you learned. Before submitting, open the uploaded work locally, confirm all filenames resolve, and compare original-file hashes with your intake record.

WEEK 3 | TASK INSTRUCTIONS

Gateway and device correlation

Objective

Correlate events without hiding uncertainty or alternative explanations.

Do this in order

1. Normalize time using a clearly stated clock-offset assumption; preserve raw device timestamps.

2. Correlate gateway reconnect events with telemetry gaps and flag matches that depend on an assumption.

3. Create a chronology with confidence and evidence references. Explain why a reconnect is not automatically evidence of tampering.

4. Give every derived timeline event an exact source file and row or event ID. Record raw time, UTC time, conversion rule, evidence meaning and uncertainty in separate fields.

5. Create adjusted_utc in a new column. For an assumed clock that is 120 seconds ahead, subtract 120 seconds; explain why that convention was selected.

6. Recheck one result independently: return to the source, repeat the calculation or lookup, and record whether it agrees. Keep a short correction log instead of silently revising previous findings.

Required hand-in

iot_timeline.csv, clock_assumptions.pdf and correlation_table.csv

Name the upload PFSA_<student-id>_iot_W3.pdf or .zip. A ZIP may contain the report plus CSV, SQL, images or other permitted supporting work. Do not rename a non-PDF file to .pdf. Keep the one portal upload below 10 MB.

Portal description and acceptance check

Write at least 200 genuine words in the description box: what you examined, how you processed it, exact evidence references, results, uncertainty and what you learned. Before submitting, open the uploaded work locally, confirm all filenames resolve, and compare original-file hashes with your intake record.

WEEK 4 | TASK INSTRUCTIONS

IoT case report

Objective

Deliver a clear, reproducible report that answers the case question.

Do this in order

1. Prepare a 4-6 page report with ecosystem diagram, telemetry quality results and correlated gateway chronology.

2. Explain the impact of clock differences, absent firmware history and unavailable physical-device context.

3. Include hashes, transformation notes, anomaly definitions and an independent-review checklist.

4. Write an executive summary first, then scope, evidence register, tool versions, reproducible methods, findings, alternatives, limitations and conclusion. Put detailed tables and queries in appendices.

5. Compare gateway disconnect/reconnect events with telemetry gaps using a documented tolerance. A timing association alone is not proof of physical tampering.

6. Recheck one result independently: return to the source, repeat the calculation or lookup, and record whether it agrees. Keep a short correction log instead of silently revising previous findings.

Required hand-in

iot_report.pdf and iot_appendix.zip

Name the upload PFSA_<student-id>_iot_W4.pdf or .zip. A ZIP may contain the report plus CSV, SQL, images or other permitted supporting work. Do not rename a non-PDF file to .pdf. Keep the one portal upload below 10 MB.

Portal description and acceptance check

Write at least 200 genuine words in the description box: what you examined, how you processed it, exact evidence references, results, uncertainty and what you learned. Before submitting, open the uploaded work locally, confirm all filenames resolve, and compare original-file hashes with your intake record.

AFTER THE WEEKLY TASK LIST

Submission & assessment

What to submit each week

Submit one main project file and a separate portal narrative of at least 200 words. Use a PDF for a readable report; use a ZIP when the report needs supporting CSV, SQL, JSON, screenshots or source code. The portal supports multiple work formats, but it is not a substitute for organising your evidence.

Item	Required content
Main report	Student ID, domain, week, case question, scope, method, findings and limitations.
Appendices	Source references, derived tables, saved formulas/queries, tool versions and relevant screenshots.
Integrity record	Original file hashes and verification result; explain whether any working file changed.
Portal narrative	At least 200 words written in your own words; a report attachment does not replace this field.

A useful 200-word narrative structure

Use approximately 40-60 words for scope and inputs, 60-90 for your actual method, 60-90 for supported findings, and 40-60 for uncertainty and learning. These are writing prompts, not text to copy. Mention at least two precise record references and one limit of the evidence. Do not repeat a sentence or paste filler just to reach the counter.

Marking rubric | 100 marks per week

Criterion	Marks	What a reviewer looks for
Integrity and scope	20	Correct baseline, preserved originals and a defined question.
Method	25	Reproducible steps, versions, calculations and saved filters.
Analysis	25	Supported reasoning and alternatives, with appropriate uncertainty.
Evidence references	20	Exact file and row/event references for each significant finding.
Communication	10	Clear report, organised package and meaningful portal description.

Calendar deadlines and any resubmission instructions are published in the portal by staff. Four submissions unlock the next document stage; payment confirmation, project assessment and document publication remain staff-controlled. Ask Support about an incorrect upload or locked record.

LAPTOP, SOFTWARE & FILES

Requirements before you begin

Laptop and workspace

Use a laptop or desktop with a supported Windows, macOS or Linux installation, a keyboard and permission to install the required applications. For these small training exports, 4 GB RAM is a practical minimum; 8 GB is recommended. Reserve at least 2 GB free for tools and working files, with additional space for downloaded books. A phone can read PDFs and check portal status, but a laptop is strongly recommended for evidence tables, file packaging and reproducible work.

Required software for this domain

LibreOffice Calc and Writer; plain-text editor; optional Python 3 for grouping and timestamps. No sensor board, soldering kit, radio adaptor, firmware tool or cloud subscription is required.

Common tools and their purpose

Tool	Use in the assignment
LibreOffice Calc	Import CSVs, retain text identifiers, calculate summaries and save derived tables.
LibreOffice Writer	Write the case report and export it as a readable PDF.
Plain-text editor / VS Code	Inspect JSON, JSONL, SQL and notes without altering formatting in a word processor.
Native SHA-256 utility	Create and compare an integrity baseline for each source file.
Browser and PDF reader	Read the guide, open references and submit work through the student portal.

Training files and folder structure

Download iot-training-data.zip from the matching domain card. Its central source files are devices.csv, telemetry.csv and gateway.csv. Save the original ZIP, extract it once, and create four folders: originals, working, analysis and submission. Put the original extracted files in originals, analysis copies in working, derived outputs in analysis, and only final hand-in material in submission.

Before the first task

Record your operating system and the installed application versions. Confirm that the file extensions are visible and that you can create a CSV, open a PDF and make a ZIP. Avoid syncing personal or confidential data into this case folder. The training pack is fictional; no purchase, live investigation account or collection from another person is needed.

PRACTICAL SETUP | WINDOWS, MACOS & LINUX

Installation & first-run checks

1 | Install a document and spreadsheet suite

Open the official LibreOffice download page and choose your operating system and CPU architecture. On Windows, download the stable main installer, run it and follow the standard setup prompts; open both Calc and Writer after installation. On macOS, choose Intel or Apple Silicon as appropriate, open the DMG and move LibreOffice to Applications. On Linux, prefer the distribution's supported software manager. Record the version in your case notes.

LibreOffice download and installation instructions

2 | Set up a plain-text editor

Download VS Code from its official site if you do not already have a suitable editor. On Windows, the user installer normally avoids a machine-wide installation. Open only your working case folder using File > Open Folder. Inspect JSON or SQL as text, keep automatic file formatting off for evidence copies, and do not add extensions that are not needed. A built-in text editor is sufficient when installing VS Code is not possible.

Official VS Code Windows setup

3 | Verify hashes without an extra package

On Windows, open PowerShell in your working case folder and run Get-FileHash -Algorithm SHA256 .\originals\<filename>. On macOS use shasum -a 256 originals/<filename>. On Linux use sha256sum originals/<filename>. Replace the placeholder with one actual filename. Save each digest with its filename, size, date and tool. Repeating the command against the same preserved file should produce the same digest.

4 | Domain-specific installation and first run

Python 3 is optional for repeatable checks. Use the official Python download page, select a supported stable Python 3 distribution and follow its operating-system instructions. Open a new terminal and verify python --version, python3 --version or py --version as appropriate. Standard csv, json, hashlib, datetime and sqlite3 modules are enough for these labs; third-party packages are not required.

Python official downloads

Installation checks: create a test worksheet, export one PDF, read one source file from a working copy and reproduce a SHA-256 digest. Save a short tools.txt with versions and any differences from these instructions. Official setup pages are the source of truth if a vendor changes a screen label.

DOMAIN-SPECIFIC GUIDANCE

Worked method & reading library

First reproducible examination

Import devices.csv and create a device-to-gateway map with each clock offset and unit. Join metadata to a working telemetry table by device_id while retaining unknown devices. Calculate gaps between consecutive readings of the same device. Compare raw and adjusted timelines to gateway.csv and mark any conclusion that changes when the offset assumption changes.

A limit you must state

This exercise does not include physical acquisition, firmware history or a live sensor. A reconnect or missing sample cannot by itself prove that someone interfered with a device.

Three books & substantial reference downloads

Use the original publisher or author links below. Reference guides and manuals are labelled separately from textbooks. Older editions provide background; they are not a substitute for current tool documentation. Third-party authors retain their rights and do not endorse PFSA.

1 | Introduction to Embedded Systems: A Cyber-Physical Systems Approach

Edward A. Lee and Sanjit A. Seshia | Second edition, MIT Press, 2017 | Author-provided textbook

Study models, time and interactions between computing and physical processes.

Open authorised PDF / author download page

2 | The Internet of Things: An Overview

Karen Rose, Scott Eldridge and Lyman Chapin / Internet Society | 2015 overview | Technical overview

Read about IoT communication models, interoperability and privacy context.

Open authorised PDF / author download page

3 | IoT Device Cybersecurity Capability Core Baseline

Michael Fagan et al. / NIST | IR 8259A, 2020 | IoT technical reference guide

Relate device identification, configuration and state awareness to an evidence inventory.

Open authorised PDF / author download page

Download links checked 13 November 2026. Use the source page if a PDF host changes or restricts automated downloads. These reading files are not required attachments to your weekly submission.

FINAL PAGE | PLAN, PRACTISE, VERIFY

Four-week learning roadmap

Your four-stage learning path

Device ecosystem map -> telemetry quality review -> clock-aware correlation -> IoT case report

Use this roadmap to plan the sequence, not to calculate calendar deadlines. Staff publish the actual dates in your portal. Early Start changes when access opens; it does not remove any weekly deliverable or assessment requirement.

Stage	Suggested focus	Checkpoint before moving on
Week 1 | Preserve	Days 1-2: scope and inventory. Days 3-4: integrity and working copy. Days 5-7: review and submission.	Every source has an ID, hash and documented limitation.
Week 2 | Examine	Days 8-9: field definitions. Days 10-11: structured checks. Days 12-14: independent recheck and submission.	Derived tables retain original IDs and reproducible calculations.
Week 3 | Correlate	Days 15-16: time conversion. Days 17-18: source correlation. Days 19-21: alternatives and submission.	Timeline retains raw time and explains ambiguity, not just a neat order.
Week 4 | Communicate	Days 22-23: findings. Days 24-25: report and appendices. Days 26-28: independent review and final hand-in.	A reviewer can repeat the method and trace every significant claim.

Daily working habit

Start by reading your last case note and checking the source hash. Work on one defined question, save the method and result, then write a brief end-of-session note: completed work, decisions, outstanding questions and next step. Keep the final report consistent with the latest version of the analysis tables.

Final readiness gate

Confirm four genuine weekly submissions, complete evidence references, original-file integrity checks and an organised final report. Read staff feedback and resolve requested corrections. Keep a personal copy of the submitted package and the portal acknowledgement. Document eligibility and recommendation remain subject to assessment and the academy's published service terms.

What you should be able to demonstrate

Explain the supplied evidence boundary, reproduce one key finding, compare a competing explanation and describe which additional evidence would improve confidence. That explanation is more valuable than a screenshot of a tool running.

Student portal and Support